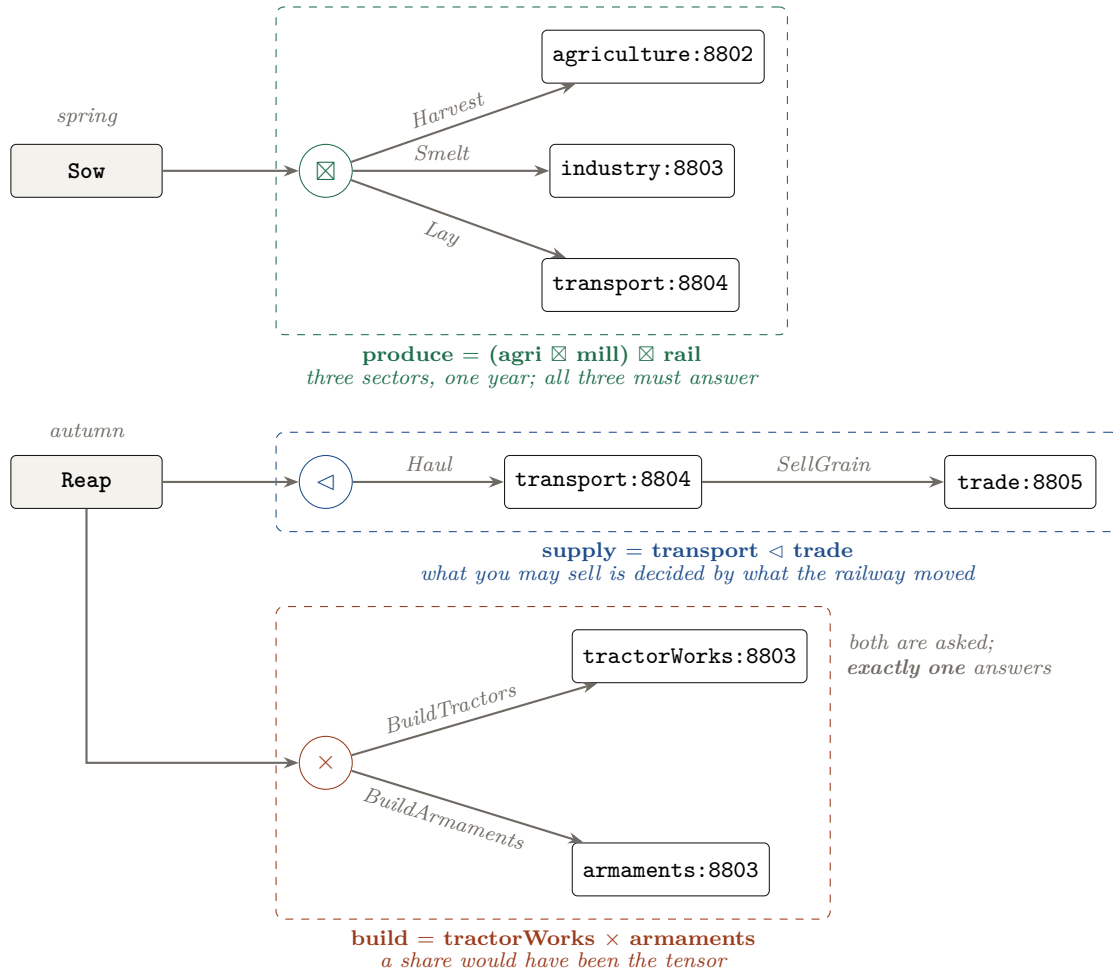


# The four combinators, and where each one fires

stalin — §5 of Ghani, *Containers for Typed Agentic AI*

A container is  $(P, R)$ : a type of prompts, and a prompt-indexed type of valid replies.

Every box below is one. Every arrow is a prompt going down and a reply coming back.



| what it means                                                                                                                                                            | where it fires in stalin                                                                                                                                                                                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ⊠ shapes multiply, positions multiply:<br>ask both, and both must answer                                                                                                 | the three sectors work the <i>same</i> year.<br><b>Promise.all</b> inside <b>tensorC</b> is not an optimisation, it is the definition.                                                                                                                                                                        |
| ◁ the shape is a dependent pair<br>( $s : A ** A.Pos s \rightarrow B$ ): a first prompt<br><i>together with</i> a function choosing<br>the second from the first's reply | what can be <i>sold</i> depends on what the<br>railway actually <i>moved</i> . The second prompt<br>is not knowable when the first is sent.                                                                                                                                                                   |
| ⊗ offer both, owe exactly one<br>— Either ( $Rp$ ) ( $Tq$ )                                                                                                              | this year's steel becomes tractors <i>or</i><br>armaments. A <i>share</i> would have been the<br>tensor; calling that a product would be false.                                                                                                                                                               |
| ⊕ routing: a coproduct of shapes,<br>eliminated by case analysis                                                                                                         | <b>never built as a composite.</b> It appears as<br>each container's own shape type — a tagged<br>union — and its eliminator is the <b>switch</b><br>in every server's dispatch. Real work, in the<br>place §3 always put it; but no <b>sumC(...)</b> call,<br>and it would be an overstatement to claim one. |

**Two that used to be here.** While the line to the port was narrow the sale was a *product* (grain *or* steel, one contract) and became a *tensor* once enough rail was laid — rail capacity choosing between  $\times$  and  $\boxtimes$ . Selling steel was removed from the game, so both would have been running on a quantity that is permanently zero. They were deleted rather than left for a diagram to claim.

**One that broke and was repaired.** Splitting the year put produce in spring and the haul in autumn, with a player decision between them, so **seqC(produce, transport)** could no longer run in a single call — it stayed declared and silently never fired. The dependent pair had not disappeared; it had moved one link down the chain. Hence **seqC(transport, trade)**: haul, *then* sell what reached the port.